W-L-111c
February 11, 1958
SUPERSEDING
Fed. Spec. W-L-111b
March 18, 1937

FEDERAL SPECIFICATION

LAMP, INCANDESCENT, (ELECTRIC, MINIATURE, TUNGSTEN-FILAMENT)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. CLASSIFICATION

1.1 Types, classes, and sizes.—All lamps covered by this specification are miniature tungsten-filament, electric incandescent lamps. "Miniature" lamps designate broadly lamps designed for automotive, flashlight, and similar types of service, and are usually equipped with bases smaller than those used on large lamps. Large tungsten-filament incandescent lamps are covered by the latest revision of Federal Specification W-L-101 and current Annual Supplement thereto.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

PPP-B-566—Boxes, Folding, Paperboard.
PPP-B-585—Boxes; Wood, Wirebound.
PPP-B-591—Boxes; Fiberboard, Wood-Cleated.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636—Boxes, Fiber.
PPP-B-676—Boxes, Set-Up, Paperboard.

Federal Standards:

Fed Std. No. 102—Preservation, Packaging, and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-B-138—Boxes, Wood, Fiberboard-Lined for Overseas Shipment (For Weight of Contents Not Exceeding 500 Pounds).
MIL-L-10547—Liners, Case, Waterproof.

Military Standards:

MIL-STD-129—Marking For Shipment and Storage.

(Copies of Military Specifications and Standards required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.—The following publication forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

Uniform Freight Classification Rules and Regulations.

(Application for copies should be addressed to Uniform Freight Classification Committee, 1 Park Avenue, New York 16, New York.)

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3. REQUIREMENTS

3.1 General requirements.—The lamps shall be so designed and made that they will satisfactorily meet the lighting-service conditions. All lamps shall conform to the manufacturer's standard shapes, sizes, and finishes of bulbs, forms of filament, and types of leading-in wires and bases on file at the National Bureau of Standards.

3.1.1 *Marking.*—The rating or trade number and the manufacturer's name or trade-mark shall be placed on the bulb or on the base of each lamp and on its container. Omission of the rating or trade number and the manufacturer's name or trade-mark where space does not permit marking on the bulb or base of the lamps shall be subject to waiver by the Federal agency for which the lamps are being procured.

3.1.2 This specification, insofar as it refers exclusively to mechanical and physical characteristics, shall apply to all miniature tungsten-filament lamps regularly listed in the manufacturers' current schedules. The full provision of this specification shall apply only to lamps specifically mentioned in the tables of initial rating tolerances, in the Annual Supplement, and to no other lamps except by mutual written agreement.

3.2 Detail requirements.

3.2.1 *Life to burn out.*—The lamps furnished under this specification shall give an average total life to burn out not less than the total life specified in the Annual Supplement, schedules of commercial ratings, or as may be modified by 4.8.6 when tested at or computed to the manufacturer's declared initial candles-per-watt.

3.2.2 *Manufacturer's declared efficiency.*—The manufacturer's declared initial candles-per-watt rating shall be not less than a value 4 percent below that shown in schedule of commercial ratings (see Annual Supplement). For the purpose of maintaining the rated life given in the Annual Supplement, the declared candles-per-watt ratings of any size of lamp may be changed at any time by notice from the manufacturer, provided that the preceding stipulation is complied with at all times.

3.2.3 *Candlepower maintenance.*—The candlepower maintenance of the lamps furnished under this specification, expressed in percent of average initial candlepower of the lamps shall equal or exceed the value specified in the Annual Supplement, schedules of commercial ratings, or as modified in 4.8.8.

3.3 *Army purchases only.*—For Army purchases the following special requirements shall apply.

3.3.1 Lamps supplied to the Army shall not fall below the minimum mortality curve shown in figure I, when tested by the Army as specified in 4.9.

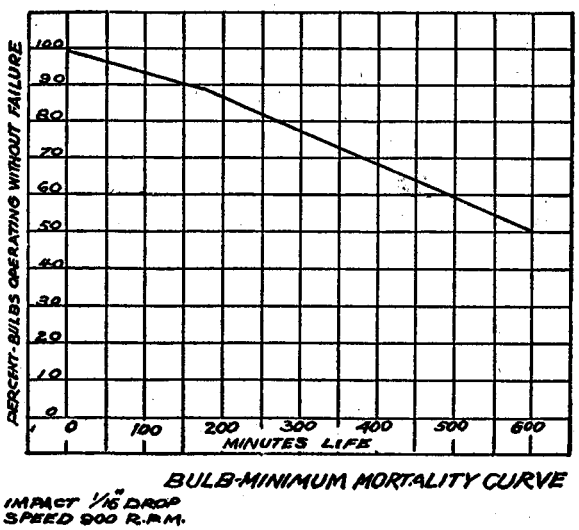


FIGURE 1.

3.4 Workmanship.—The manufacturer is given the widest range in the selection of materials and processes of manufacture in order that miniature tungsten-filament incandescent electric lamps of maximum quality may be produced.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Test procedure.—All tests shall be made in a competent and expert engineering manner at the expense of the purchaser, excepting that when initial tests and inspections are made at the factory, the manufacturer will be required to supply the necessary equipment, assistance, current, and facilities for making such initial tests and inspections. The manufacturer shall have the privilege of witnessing and verifying all tests of his lamps made hereunder, and shall also have the privilege of obtaining copies of

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the results of the test of his lamps and of having access to the records of such tests at all reasonable times. The method of test procedure may be modified in any particular, provided that it is mutually agreed that such modified procedure will yield results equivalent to those obtained with the procedure specified herein.

4.2 Lamps shall not be submitted for inspection within 120 hours from the time that they are manufactured.

4.3 A lot shall be all the lamps of any one type, size, and voltage offered for acceptance at any one time.

4.4 Physical inspection test.

4.4.1 *Lamp selection and test quantity.*—The number of lamps to be selected by the inspector for physical inspection shall be not less than 5 percent (minimum 20 lamps) of the lot taken at random and proportionately from the individual packages of the lot. Lamps thus selected for inspection are known as the "inspection-test lamps."

4.4.2 *Rejection for minor physical defects.*—Minor physical defects are those which mar the appearance of the lamps or are otherwise incompatible with good workmanship or good service. A lot may be rejected without further test if 10 percent plus 1 lamp (20 percent plus 1 lamp when tested elsewhere than at the factory) or more of the inspection-test lamps have physical defects.

4.4.3 *Rejection for major physical defects.*—Major physical defects are those which render lamps inoperative or which would cause them unquestionably to give poor service. A lot may be rejected without further test if 1 percent plus 2 lamps (2 percent plus 3 lamps when tested elsewhere than at the factory) or more of the inspection-test lamps have major physical defects. Individual lamps having major physical defects may be rejected and eliminated from an accepted lot and shall be replaced by the manufacturer without additional charge.

4.5 Physical dimension inspection test of automobile headlight lamps.

4.5.1 *Rejection for incorrect physical dimensions.*—A lot of lamps may be rejected for incorrect physical dimensions if 2 percent plus 1 lamp or more of the inspection-test lamps depart by more than the tolerance given in the tables of the Annual Supplement from any one of the standard dimensions as given in the schedules of the Annual Supplement. The physical dimensions are to be measured with the filaments of the lamps operated at design voltage.

4.5.2 *Physical dimensions of automobile headlight lamps with 2-pin bayonet candelabra bases.*—The measurement of certain physical dimensions of automobile headlight lamps fitted with 2-pin bayonet candelabra bases shall be referred to three planes:

Plane A is normal to the base axis and passes through a contact point of either base-locking pin which is nearest to the bulb.

Plane B passes through the base axis and a contact point of either base-locking pin which is nearest to the bulb.

Plane C passes through the base axis and is normal to plane B.

NOTE.—The contact point of either base-locking pin which is nearest the bulb is such a point which is nearest a plane intersecting the glass bulb and perpendicular to the base axis.

4.5.2.1 The center of the light source referred to in these specifications and pertaining to 2-pin bayonet base lamps, shall mean the center of the smallest rectangle, the normal projections of which on planes B and C enclose the projections of the light source on the same planes, and which has two sides parallel to plane B and C and the other two sides parallel to plane A.

4.5.2.2 The light center length shall be the distance between plane A and the center of the light source.

4.5.3 *Physical dimensions of prefocused automobile headlight lamps.*—The measurement of certain physical dimensions of prefocused automobile headlight lamps shall be referred to three planes:

Plane A is tangent to the outer surfaces of the three protruding locating bosses on the lower face (away from bulb) of the base-focusing flange.

Plane B is perpendicular to plane A and passes through the center point of the base flange and through the center of the major locking eyelet of the base flange. (The major eyelet is the one which is equidistant from the other two base-locking eyelets.)

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Plane C is perpendicular to planes A and B and passes through the center point of the base flange.

4.6 Definitions.

4.6.1 *Light center for C-2 shaped filaments.*—Light center is a point lying in a plane through the axis of the filament coils midway between the "top" and "bottom" of the lighted filament and midway between the outer edges of the filament coils at the horizontal level of the light center.

NOTE.—Filament top is the upper extremity of the highest lighted turn (away from base) in either coil segment. Filament bottom is the underside of the lowest lighted turn (toward base) in either coil segment. The lowest lighted turn is considered for this purpose as being the first turn above the first inside angle between turns of the projected filament image.

4.6.2 *Light center for C-6 shaped filaments.*—Light center is a point lying in the axis of the filament coil midway between the end turns of the lighted filament.

4.6.3 *Light center length.*—Light center length is the distance from light center to plane A.

4.7 Initial rating test.

4.7.1 *Lamp selection and test quantity.*—The number of lamps selected for rating shall be not less than 5 percent (minimum 20 lamps) of the lot, taken at random and proportionately from the individual packages of the lot. The lamps may be the same as those selected for physical inspection (4.4.1) and are known as the "rating-test lamps." The rating test lamps must have been sufficiently burned or seasoned to have reached stable values of watts and candlepower before the initial rating test is made.

4.7.2 *Rejection for spread in rating.*—Rating-test lamps shall be operated at design volts. A lot of lamps may be rejected if 20 percent plus 2 lamps (30 percent plus 2 lamps when tested elsewhere than at the factory) or more of the rating-test lamps depart from the manufacturer's declared initial rating by more than the initial rating tolerance given in the Annual Supplement, such rating tolerance to be applied to the manufacturer's declared initial rating in effect at the time of manufacture of the lamps, subject to the requirements of 3.2.2. Individual rating-test lamps may be rejected and eliminated from an accepted lot if they fall outside these tolerances by more than an additional 8 percent, and shall be replaced by the manufacturer without additional charge.

4.8 Life-performance test.

4.8.1 *Lamp selection and test quantities.*—The number of lamps for life-performance test shall in no case be less than 10 lamps and shall be selected from the rating-test lamps at the ratio of not less than 8 from the first 1,000 lamps of any lot and at least 2 lamps from each 500 lamps (or part thereof) by which the lot exceeds 1,000 lamps. The life-test lamps selected shall be representative of the rating-test lamps.

4.8.2 *Life-test voltages.*—Life-test lamps shall be operated on the test racks at design volts or at voltages corresponding either to the initial rated candles per watt as given in the Schedules of Commercial Ratings in the Annual Supplement, or at an efficiency or voltage mutually agreed upon. Operation at approximately the voltages indicated is sanctioned, provided that proper life adjustments are made.

4.8.3 *Position of burning.*—Lamps designed for a particular burning position shall be so tested. All other lamps shall be tested in the position in which they are most generally used in service.

4.8.4 *Voltage regulation.*—Accurate recording meter records shall be obtained throughout the tests to show the variation of the circuit. Variations of voltage shall not exceed one-half of 1 percent above and below the test voltage.

4.8.5 *Lamp life.*—The life of a lamp is the number of hours' life of the lamp to burn out.

4.8.6 *Rejection for failure to meet average life requirements.*—Any lot of lamps, provided that such lot is represented on test by at least the minimum required number of lamps in accordance with paragraph 4.8.1, may be rejected if the average hours' life of the life-test lamps, when computed to the manufacturer's declared initial candles per watt rating falls below the life value given in the current Annual Supplement by more than the life tolerance specified in table I, life tolerances, for the number of life-test lamps averaged, as modified by the "multiplying factor" specified in the Annual

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Supplement, in accordance with footnote 2 under table I. Such a lot of lamps may be rejected if 10 percent plus 2 lamps or more of the life-test lamps have lives less than 40 percent of rated life.

TABLE I.—*Life tolerances* [1] [2]

Number of lamps averaged	Allowable percent variation from rated life
100 and above	15
40–99	16
20–39	17
10–19	18

[1] On account of the natural and inherent variations in individual lamp performance, the average test results of a small number of test samples cannot be depended upon to indicate exactly the average for the larger quantity from which the test samples were taken. The tables of allowable variations are devised in accordance with the observed accuracy of lamp-testing methods and the observed variations in lamp performance.

[2] The life tolerance for any type of lamp is determined by multiplying the allowable percent variation in table I by the "multiplying factor" for the particular lamp, as given in the Annual Supplement.

4.8.6.1 *Life test of lamps for flashlight service.*—As it is impractical in the present stage of the art to photometer lamps for flashlight service, with the necessary degree of accuracy for rating and life test, these lamps shall be life tested at the design volts given in the Schedules of Commercial Ratings in the Annual Supplement. It should be noted that the life test results obtained on flashlight lamps are not necessarily indicative of their quality, as their efficiency is not considered. By designing the lamps to operate at low efficiencies, long-life results can be obtained, but such long-lived lamps are as undesirable, on account of the high battery consumption for the light obtained, as are normally short-lived lamps on account of the high lamp consumption for the light obtained.

4.8.7 *Candlepower maintenance of lamps.*—Life-test lamps for automotive service shall be measured for spherical candlepower at their design volts, both initially and at the expiration of four-fifths of the rated life. The measure of candlepower maintenance for an individual lamp shall be its spherical candlepower at four-fifths of the rated life, expressed as a percentage of its initial spherical candlepower.

4.8.8 *Rejection for failure to meet candlepower maintenance requirements.*—When tested in accordance with the foregoing test specifications, any group of lamps may be rejected, provided such group is represented on life-performance test by the minimum required number of lamps in accordance with paragraph 4.8.1, and provided the percent spherical candlepower at four-fifths rated life, as above derived, falls below the percent specified in the Annual Supplement by more than the tolerance specified in table II.

TABLE II.—*Candlepower maintenance tolerance* [1]

Number of lamps tested	Allowable variation in candlepower at four-fifths rated life in percent of average initial candlepower
100 and above	2
40–99	4
20–39	6
10–19	8

[1] On account of the natural and inherent variations in individual lamp performance, the average test results of a small number of test samples cannot be depended upon to indicate exactly the average for the larger quantity from which the test samples were taken. The tables of allowable variations are devised in accordance with the observed accuracy of lamp-testing methods and the observed variations in lamp performance.

4.8.9 *Life-test lamps to be counted as part of quantity delivered.*—The life-test lamps selected by the inspector will be destroyed in the course of the life test. The inspector will place a card in each container from which a life-test lamp has been taken, stating: "The United States Government lamp inspector has removed the lamp from this carton as part of the inspection of lamps in this package, and has sent the lamp to the National Bureau of Standards,

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Washington, D. C., for the required life test." Lamps destroyed in life tests are to be counted as part of the quantity of lamps delivered by the contractor to the United States Government.

4.9 Impact test.—(Army purchases only.)

4.9.1 Twenty lamps designated as impact-test lamps, shall be installed in an impact tester similar to that shown in figure 2, and tested as follows: (See 3.3).

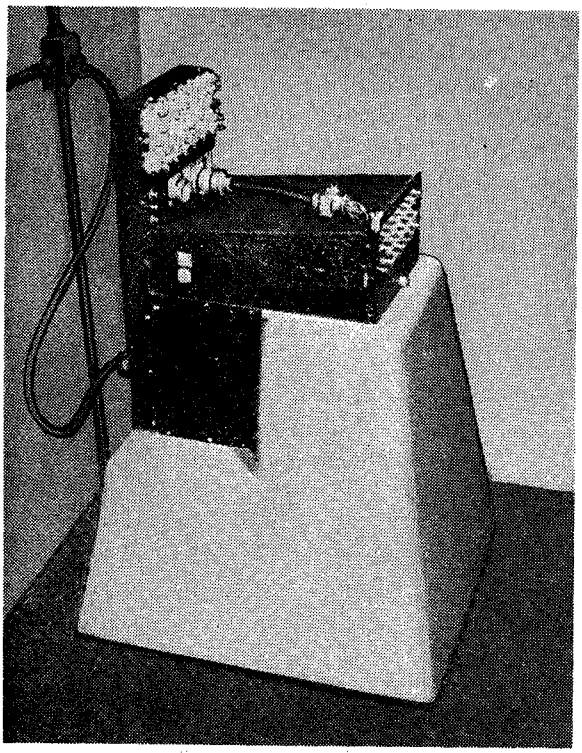


FIGURE 2.

(*a*) Test cycle for single filament lamps:
- (1) Vibrate for 15 minutes with lamps unlighted.
- (2) Vibrate for 30 minutes with lamps operating at design volts.
- (3) Vibrate for 15 minutes with lamps unlighted.
- (4) Repeat the above cycle for 6 hours or until 12 lamps fail.

(*b*) Test cycle for two-filament lamps:
- (1) Vibrate for 15 minutes with lamps unlighted.
- (2) Vibrate for 25 minutes with minor filaments operating at design volts.
- (3) Vibrate for 5 minutes with minor and major filaments operating at design volts.
- (4) Repeat the above cycle for 6 hours or until 12 lamps fail.

4.9.2 The twenty impact-test lamps referred to in 4.9.1 shall be in addition to the life-test lamps selected in accordance with 4.8.1. The impact-test lamps shall be forwarded to the laboratory specified by the Army.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 *Level A.*—The lamp shall be unitized by placing in a carton or set-up box conforming to Federal Specification PPP–B–566, or PPP–B–676, respectively, and mechanically protected by using separators, trays, or cushioning material. Not more than 25 lamps shall be packaged per unit pack. The lamps shall be further protected by placing the unit in an intermediate container conforming to Federal Specification PPP–B–636, type, class, and grade optional. The intermediate container shall contain no more than 500 lamps.

5.1.2 *Level C.*—The lamps shall be packaged in accordance with manufacturer's commercial practice.

5.2 Packing.—The lamps, packaged Level A or C, shall be packed Level A, B, or C, as specified in the contract or order.

5.2.1 *Level A.*—Lamps shall be packed in boxes conforming to the overseas requirements in accordance with Federal Specification PPP–B–585, class 3, PPP–B–591, PPP–B–601, PPP–B–621, and PPP–B–636, types I and II, classes 2 or 3, grade 3, and Military Specification MIL–B–138. Gross weight of wood and wood-cleated boxes shall not exceed 200 pounds. Case liners conforming to Military Specification MIL–L–10547, class 2 shall be used except when boxes conforming to Federal Specification PPP–B–636, classes 2 or 3 are used as interior containers. Closures shall be in accordance with the applicable box specification.

5.2.2 *Level B.*—Lamps shall be packed in boxes conforming to the domestic requirements of Federal Specifications PPP–B–585, PPP–B–591, PPP–B–601, PPP–B–621, and PPP–B–636, types I or II, class 1. Gross weight of wood and wood-cleated boxes shall not exceed

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200 pounds. Closures shall be in accordance with the applicable box specification.

5.2.3 *Level C.*—Lamps shall be packed in containers of the type, size, and kind commonly used for the purpose, in a manner that will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the Uniform Freight Classification Rules, or regulations of other carriers as applicable to the mode of transportation.

5.3 Marking for shipment.—In addition to any special marking required in the contract or order, marking of the shipping containers shall be in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging and packing protection for supplies and equipment.

6.2 Ordering data.—Purchasers should specify size and kind of lamp required and level of packing and packaging desired. (See Section 5.)

6.3 Qualifying to supply lamps.

6.3.1 *Current contractors.*—A contractor currently supplying miniature lamps to the Government must qualify as follows in order to have his bid for the contract period year considered: The weighted average life of all of said contractor's miniature lamps whose life test is completed by the National Bureau of Standards during the first 10 months of the current contract period, both at the manufacturer's declared initial efficiency and at the initial efficiency as listed in the current supplement to this specification when expressed in percent of the rated lives of all of the sizes of lamps tested shall not fall below 100 with a life tolerance for the total number of lamps tested as listed in table I. The weighted average candlepower maintenance, as defined in paragraph 4.8.7 of the lamps tested, shall not fall below the weighted average of the candlepower maintenance values specified in the Annual Supplement to this specification, with applicable tolerances.

6.3.1.1 The bid for the ensuing contract period of a contractor currently supplying miniature lamps to the Government will not be considered if 14 percent or more of all of the lots of miniature lamps of schedule 2 of said contractor's manufacture whose life-test is completed during the first 10 months of the current contract period fail to comply with the requirements of paragraph 4.8.6 (with the words "the initial candles-per-watt given in schedule 2 of the current Annual Supplement" substituted for "the manufacturer's declared initial candles per watt rating") or paragraph 4.8.8 or both of this specification.

6.3.1.2 The requirements in 6.3.1 and 6.3.1.1 apply to any current contractor only insofar as he supplies lamps of his own manufacture. If a current contractor supplies one or more brands of lamps not of his own manufacture, such brand or brands shall be considered as part of the qualification test or tests of the manufacturer or manufacturers of such brand or brands.

6.3.2 *Prospective bidders.*—A prospective bidder not currently supplying miniature lamps to the Government on contract must qualify as follows in order to have his bid for the ensuing contract period considered.

6.3.2.1 Upon notice from the prospective bidder, not less than 5 months before the expiration of the current contract period, to the Federal Supply Service of his intention to bid to supply miniature lamps on contract during the ensuing contract period, the Federal Supply Service will notify the National Bureau of Standards to make a qualification test of the prospective bidder's product. The National Bureau of Standards will notify the prospective bidder to have ready for inspection by a Government lamp inspector, 2,000 of each of eight of the following automotive service lamps as listed in schedule 2 of the current Annual Supplement: trade numbers 51, 55, 63, 82, 87, 1000, 1129, 1133, 1142, and 1158.

6.3.2.1.1 The procedure specified in 6.3.2.1 applies only to any prospective bidder who contemplates supplying lamps of his own manufacture. If a prospective bidder contemplates supplying lamps not of his own manufacture, the manufacturer of the lamps, if not already undergoing qualification tests as a current contractor, shall give notice to the Federal Supply

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Service as required in 6.3.2.1 as though he were the prospective bidder. If the manufacturer of the lamps is already undergoing qualification tests as a current contractor, no further qualification test is required.

6.3.2.2 The Government lamp inspector will inspect the lamps as provided in 4.1, 4.2, 4.4.1 and 4.7.1 of the specification. The prospective bidder shall declare the initial efficiency of each type of lamp submitted as provided in paragraph 3.2.2. The Government lamp inspector will select at random 14 of the rating-test lamps of each size presented for inspection, and will send the lamps thus selected to the National Bureau of Standards for photometric rating and life test. These life-test lamps shall be supplied by the prospective bidder without charge and all transportation charges shall be prepaid by the prospective bidder. The Government lamp inspector will at the same time inspect the facilities of the prospective bidder to determine whether they are suitable and sufficient to enable him to efficiently inspect lamps at the factory or warehouse of the prospective bidder, if the prospective bidder should be awarded a contract to supply miniature lamps to the Government during the ensuing contract period.

6.3.2.3 The bid of the prospective bidder for the ensuing contract period will not be considered by the Federal Supply Service, if, upon measurement at the National Bureau of Standards, 10 percent or more of the lamps depart from any of the dimensions specified herein by more than the allowable tolerances.

6.3.2.4 The lamps selected by the Government lamp inspector for life test shall be photometered at the National Bureau of Standards. If 20 percent plus 2 lamps or more of all of these lamps are outside of the initial rating tolerances, the bid for the ensuing contract period of the prospective bidder will not be considered by the Federal Supply Service.

6.3.2.5 The life test of the various sizes of lamps selected by the Government lamp inspector shall be conducted at the National Bureau of Standards in accordance with the provisions of 4.8.2, 4.8.3, 4.8.4, and 4.8.7 of this specification. The Federal Supply Service will consider the bid of the prospective bidder for the ensuing contract period only if the weighted average life, both at the manufacturer's declared initial efficiency listed in the Annual Supplement to the current issue of Federal Specification W–L–111 when expressed in percent of the rated lives of all of the sizes of lamps tested, does not fall below 100, with a life tolerance for the total number of lamps tested as listed in Table I.

6.3.2.6 The Federal Supply Service will consider the bid of the prospective bidder for the ensuing contract period only if the weighted average candlepower maintenance as defined in 4.8.7 of the lamps tested does not fall below the weighted average of the candlepower maintenance values specified in the Annual Supplement to this specification with applicable tolerances.

6.3.2.7 The Federal Supply Service will not consider the bid of the prospective bidder if 14 percent or more of the total number of lots of the prospective bidders manufacture which are life-tested, fail to comply with the requirements of paragraph 4.8.6 (with the words "the initial candles per watt in the current Annual Supplement" substituted for "the manufacturer's declared initial candles per watt rating") or 4.8.8, or both, of this specification.

6.3.3 Only one qualification test of any one manufacturer's lamp shall be made in any one contract year.

6.4 This specification has been prepared primarily for the use of the departments of the Government in purchasing incandescent lamps. In circulating it, however, attention is directed to the necessity of extreme caution in the application of the tests described. Only those thoroughly instructed in the science of photometry and experienced in life testing of lamps should undertake to determine the acceptability of lamps under the terms of this specification.

6.5 Definitions and standards.

6.5.1 *Electrical units.*—The electrical units referred to in this specification are the international units which have been in force since January 1, 1911.

6.5.2 *Unit of luminous intensity.*—The unit of luminous intensity shall be the international candle as maintained by the National Bureau of Standards at Washington, D. C.

6.5.3 *Photometric measure.*—The basis of photometric measure for all lamps shall be the

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average luminous intensity expressed in spherical candlepower.

6.5.4 The (mean) spherical candlepower of a lamp is the average candlepower of the lamp in all directions in space. It is equal to the total luminous flux of the lamp divided by 4π.

6.5.5 *Style of lamps.*—Lamps are classified under two styles, large and miniature. All lamps covered by this specification are miniature lamps.

6.5.6 *Types of lamps.*—The word "type" is used to distinguish between lamps which differ in general construction or as designed for different classes of service.

6.5.7 *Rating of lamps.*

6.5.7.1 *Size of lamps.*—The size of incandescent lamps is expressed in rated watts or rated candlepower.

6.5.7.2 *Rated volts.*—The rated volts of a lamp is the voltage for which the lamp is designed.

6.5.8 *Design and size of bulbs.*—The design of a bulb is designated by a letter. The size of a bulb is expressed by a number denoting the greatest diameter in eights of an inch.

6.5.9 *Regular lamps.*—Regular lamps are lamps whose construction conforms to that regarded as standard and are so listed by the manufacturer. Only miniature regular lamps are covered by this specification.

6.6 Lamps which have not been used and are rejected under the terms of this specification may be returned to the manufacturer at his expense and no payment made therefor. All lamps placed in service shall be considered as accepted.

6.7 Transportation description.— Transportation description applicable to these items is:

Lamps, Electric, Incandescent, including Photo-flash.
Carload minimum weight 15,000 and 24,000 pounds, subject to Rule 34, Uniform Freight Classification.
Motor volume minimum weight—any quantity.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—O E M T
Navy—Sh A MC Or S
Air Force

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